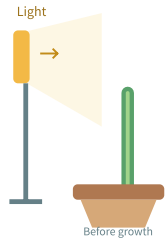
1. Complete the treatment labels

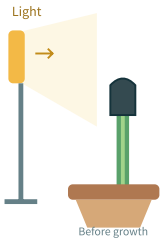
Choose labels: uncovered, tip shaded, shaded below the tip. All tips are intact and the light is on the left.

A



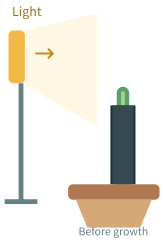
Treatment label:

B



Treatment label:

C



Treatment label:

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.

2. Variables and assumptions in the main comparison

Independent variable(the factor changed on purpose)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

Dependent variable(the result measured or observed)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

Controlled variables(the factors kept the same)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

03 • Experiment assumptions

What assumptions does this inquiry make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all suitable assumptions. You may choose more than one.

- ☐ The coleoptiles（胚芽鞘） start in similar growth conditions.
- ☐ The light-blocking cap and sleeve do not limit elongation（延長） or bending.
- ☐ The treatments do not cause clear temperature differences between setups.
- ☐ If the plants are the same type, their starting size does not need to be controlled.

3. What baseline does uncovered A provide? What alone changes in A–B and A–C? Give at least two conditions that must be the same.

(Hint: The control group provides a baseline for comparison. It helps you confirm that a change is caused by the independent variable rather than other external factors, so the control group does not receive the independent variable treatment.)

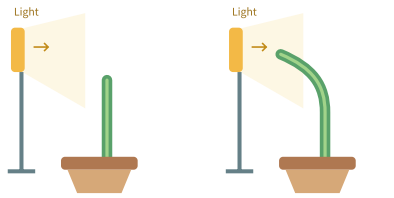
4. Observe and analyse shading comparisons

Light from the same side, three light-blocking treatments

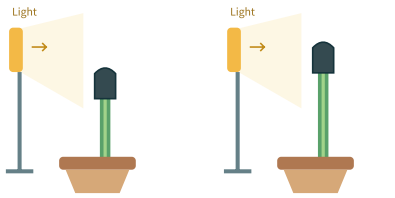
The light is on the left for every setup. Before treatment, the coleoptiles（胚芽鞘） are similar in type, starting height and growth stage. Light intensity, exposure time, temperature and water supply are the same. The opaque（不透光） sleeve covers the whole part below the tip, leaving the tip exposed. The sleeve can bend as the plant grows and does not limit elongation（延長）.

The animation speeds up growth and shows a typical response. The time and data are teaching simulations, not real measurements. The model does not show differences between individual plants.

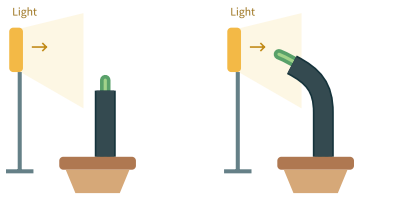
A • No light blocked



B • Light blocked at the tip



C • Light blocked below the tip



Setup	Treatment	Bending direction
A	No light blocked	（Bends left ／ Bends right ／ No clear bending）
B	Light blocked at the tip	（Bends left ／ Bends right ／ No clear bending）
C	Light blocked below the tip	（Bends left ／ Bends right ／ No clear bending）

Hint: compare treatments and before/after shapes, using A as the baseline.

1. Compare A and B: after **blocking light at the tip**, what changes?（No clear bending towards the light. ／ It shows the same bending towards light.）

2. Compare A and C: when the lower part **does not** receive light directly, does the coleoptile（胚芽鞘） still bend?（It can still bend towards light. ／ No clear bending towards the light.）

3. Which part may **sense light from one side**?（Tip ／ Only the lower part）

5. Compare elongation（延長） of intact A and cut-tip D

Use the same left-side light and incubation conditions; compare whether the tip is present. Use the before/after images to circle yes/no elongation（延長）, then complete the conclusion. Cutting also creates a wound; do not infer a unique mechanism.

A • No light blocked



D • Tip removed



Setup	Treatment	Elongation（延長）
A	No light blocked	（Yes ／ No）
D	Tip removed	（Yes ／ No）

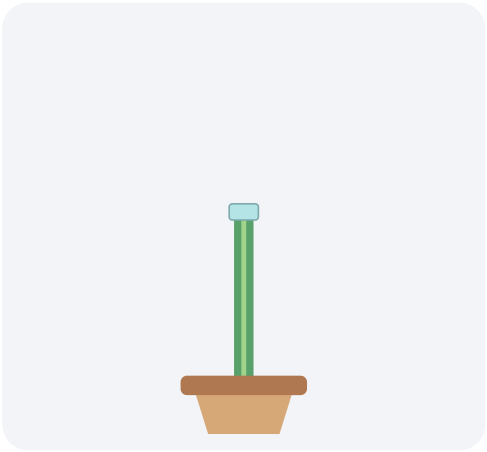
1. What does comparing elongation（延長） in A and D support?（The presence of the tip also affects the coleoptile（胚芽鞘）’s growth / elongation（延長）. ／ The presence of the tip does not affect the coleoptile（胚芽鞘）’s growth / elongation（延長）.）

6. Record E/F angles and directions in darkness

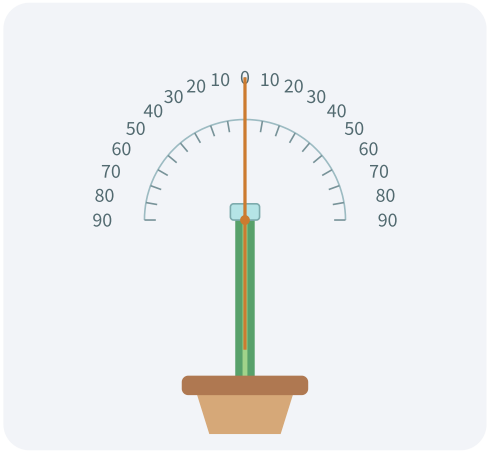
All E–H are incubated in darkness for 24 hours (a teaching model), with the same species, initial size, time, temperature and water. Use the A2 method to read final protractors: angle is the magnitude from vertical; record direction separately. Agar（瓊脂） type and position remain fixed during incubation.

Extension 2 materials: oat coleoptiles（胚芽鞘） with tips removed × 4, growing containers × 4, blank agar（瓊脂） × 1, treated agar（瓊脂） × 3, timer × 1.

E • Blank agar（瓊脂） • Centre

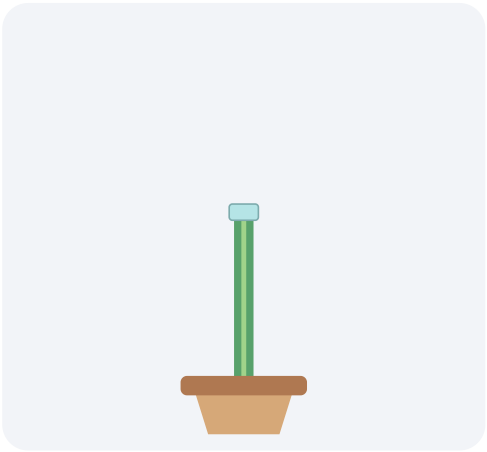


Darkness • Before growth

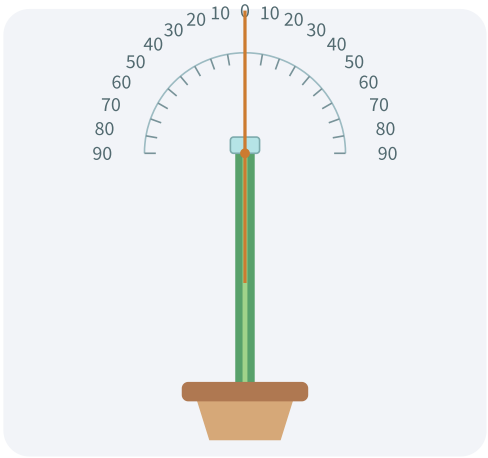


Darkness • After growth

F • Treated agar（瓊脂） • Centre



Darkness • Before growth



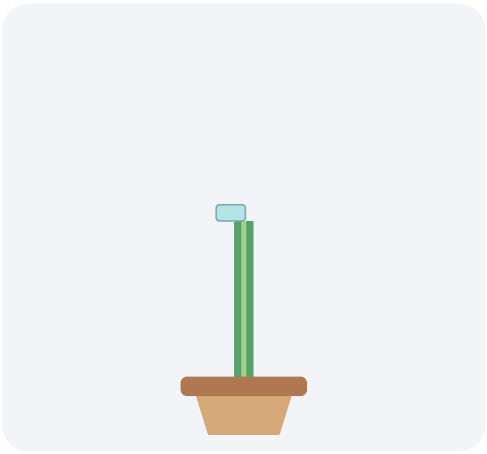
Darkness • After growth

Setup	Treatment	Angle (°)	Bending direction
E	Blank agar（瓊脂） • Centre	_____°	（Bends left ／ Bends right ／ No clear bending）
F	Treated agar（瓊脂） • Centre	_____°	（Bends left ／ Bends right ／ No clear bending）

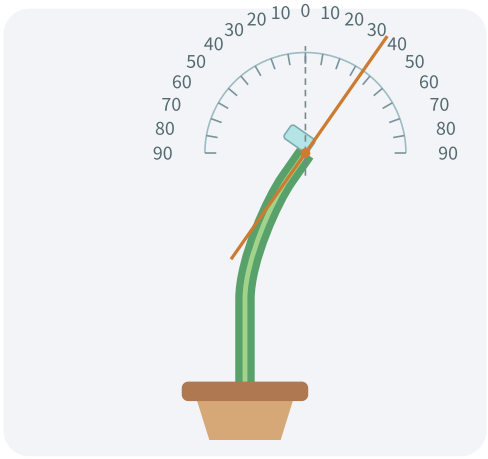
6. (continued) Record G/H and analyse E–H

G/H are also incubated in darkness. They use the same treated agar（瓊脂） at different positions. Read the angle and record direction, then answer the three questions below.

G • Treated agar（瓊脂） • Left

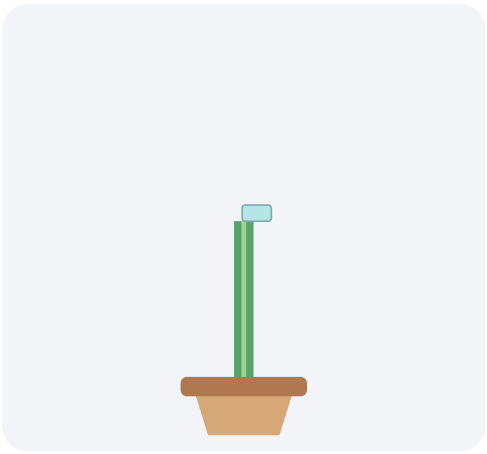


Darkness • Before growth

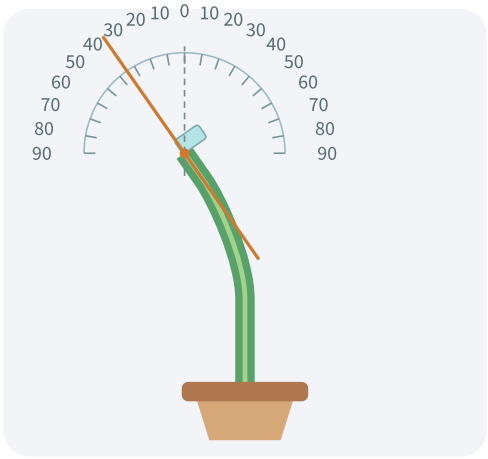


Darkness • After growth

H • Treated agar（瓊脂） • Right



Darkness • Before growth



Darkness • After growth

Setup	Treatment	Angle (°)	Bending direction
G	Treated agar（瓊脂） • Left	_____ °	（Bends left ／ Bends right ／ No clear bending）
H	Treated agar（瓊脂） • Right	_____ °	（Bends left ／ Bends right ／ No clear bending）

1. What does the comparison of E and F support?（The tip may produce a growth-promoting effect that can be transferred. ／ All agar（瓊脂） produces the same growth-promoting effect.）

2. What does the comparison of F, G and H support?

◯ The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending.

◯ The position of treated agar（瓊脂） **does not** affect the direction of coleoptile（胚芽鞘） bending.

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives （more ／ less） substance X, causing cells on the left to **elongate（延長）** （more ／ less） and the coleoptile（胚芽鞘） to bend to the （left ／ right） side.

7. Identify independent variables in different comparisons

Circle one option in each row; distinguish variables across the investigations.

A–D:

（whether the tip is present ／ light direction）

E–F:

（whether agar（瓊脂） previously touched the tip ／ agar（瓊脂） position）

F–G–H:

（whether agar（瓊脂） previously touched the tip ／ treated agar（瓊脂） position）

8. Observe the window-side context, then apply the concepts

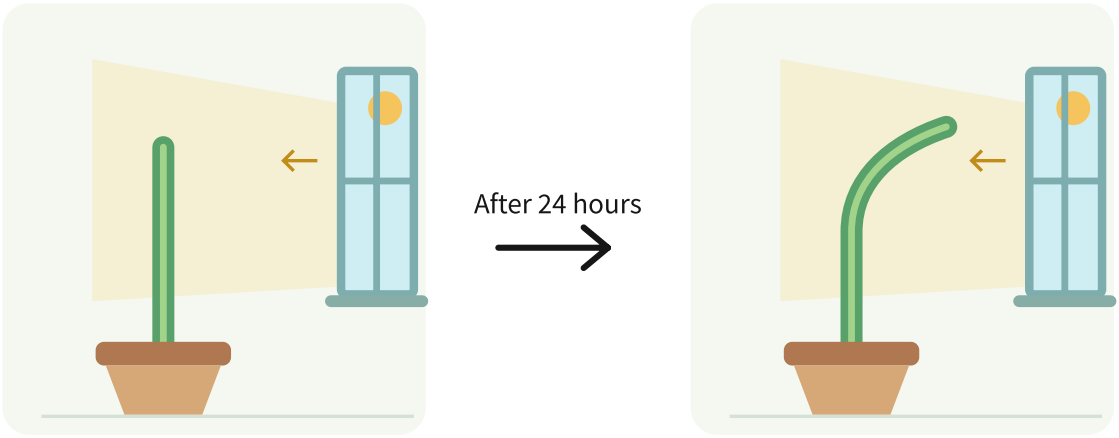


Illustration of growth changes; 24 hours is the time chosen for this situation.

Describe the change in coleoptile（胚芽鞘） shape and growth direction in the before/after images.

Use the learning points to explain why the coleoptile（胚芽鞘） in the diagram bends to the right.

Connect light direction, the shaded side, auxin（生長素） and differential elongation（延長） of cells on the two sides.

Keep these first answers. Read C only after completing B; write corrections separately without erasing the originals.